

Variable Filter Design For Grid-connected Solar Inverter Systems



Presented by:
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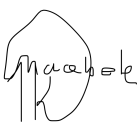
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Acknowledgments

Firstly, I would like to thank the Lord Almighty for giving me the grace to complete this project. I want to also express my gratitude to my supervisor, Dr Sampath for his invaluable knowledge, patience and guidance throughout my project work.

Abstract

LCL filter designs are extensively documented in the literature, with a design process that typically follows an iterative approach due to the numerous constraints on filter parameters and adherence to IEEE519 guidelines for permissible grid-injected current ripple. Traditional design methodologies for LCL filter components—such as the inverter-side inductor, grid-side inductor, and filter capacitor—often involve maintaining fixed values. This approach fails to account for the variations in load or temperature conditions within the system.

This project proposes a variable filter design methodology to optimize the selection of inverter-side and grid-side inductance, thereby satisfying design requirements across diverse applications. These applications include minimizing inductor core volume, reducing power losses, improving total harmonic distortion (THD), and managing temperature variations. The proposed design approach ensures a THD of less than 5%, complying with the IEEE519 standard for all design scenarios.

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Chapter 1

Introduction

1.1 Background to the study

The global transition towards renewable energy sources, such as solar, wind and fuel energy, has led to an increased need for effective integration of power filter into conventional electrical grid networks [2], [9]. Central to this integration is the use of grid-connected inverters, which convert the direct current (DC) generated by renewable energy sources into alternating current (AC) injected into the grid [11]. However, the power quality issues introduced by the inverters due to pulse-width modulation (PWM) utilisation techniques can disrupt grid stability and efficiency [13].

Some of these power quality issues includes harmonics due to switching in inverters are distortions in the electrical waveform, voltage sags and transients [13]. Harmonics can cause overheating in the electrical equipment, reduce the efficiency of power transmission, and lead to malfunctioning of sensitive electronic devices. Voltage sags and transients can result in power outages and equipment damage, posing significant challenges to maintaining a stable and reliable power supply.

To address these issues, filters are employed in grid-connected inverter systems to smooth out the inverter output currents and improve the quality of the power delivered to the grid [6]. Traditional filter designs, such as a simple L-filter have been extensively used to mitigate harmonics and other power quality problems. But these filters often results in bulky design of the inductor and poor harmonic attenuation [17]. In most conventional filter designs, the filter parameters are usually fixed which limits the performance of the filter under varying temperatures and dynamic load changes.

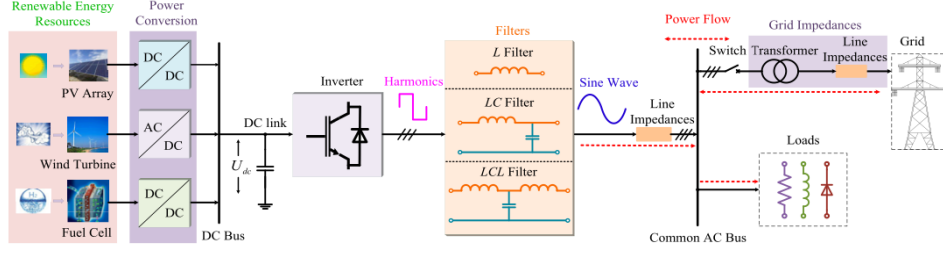


Figure 1.1: Renewable energy sources with grid-connected inverter system [2]

The growing interest in variable filter design is driven by the need for more efficient and resilient power systems that can accommodate the increasing penetration of renewable energy sources. By improving power quality and stability, variable filters not only enhance the performance of grid-connected inverters but also contribute to the broader goals of energy sustainability and reliability.

This study aims to explore different filter topologies, state-of-the-art filter designs for grid-connected inverter systems identifying the key methodologies and challenges. By providing a detailed understanding of the existing literature, this research will contribute to the development of an optimal variable filter that will be used to reduce the weight and size of the filter, reduce the power losses and improve the overall performance of the filter.

1.2 Objectives of this study

The objectives of the project are:

1. To review the literature on state-of-the-art filter designs for grid-connected inverters that utilizes magnetic properties.
2. To develop a methodology to design filters for grid-connected inverters by utilizing the modern magnetic materials.
3. To verify the efficacy of the filter using MATLAB Simulink

1.2.1 Purpose of the study

Due to an increase in the utilisation of renewable energy sources in modern power system, there is a need to ensure seamless integration of distributed energy sources with grid-connected inverter systems to supply quality power into the grid networks [13]. Because of this, there's a need to investigate different filter designs that will improve the efficiency, power quality, cost as well as the overall weight and size of the filter components.

1.3 Scope and Limitations

This project, with a 14-week deadline, is structured into three key phases to ensure a focused and timely completion of the project report: The first deliverable was reviewing different filter designs for grid-connected solar inverter systems and then, come up with an optimal variable filter design that will be used to select the best inverter-side and grid-side inductance combinations to achieve the best performance. Lastly, the utilisation of Matlab to verify the filter design performance under certain requirements of the system.

1.4 Plan of development

This project is organized in 7 chapters.

Chapter 1: Introduction

This chapter describe the background to the study, objectives, purpose, scope and limitations as well as the plan of development of the project.

Chapter 2 - Literature Review

This chapter presents a review of different filter topologies (L,LC,LCL-filters) for grid-connected inverter systems, and state-of-the art filter design methodologies incorporating soft magnetic properties. Firstly, it distinguishes the filter topologies and then, focus more on LCL-filters to determine the best methodology that is used for calculating the filter parameters.

Chapter 3: Implemented LCL-filter Design Methodology

This chapter details the variable filter design methodology used in this project. The

Matlab Simulink model and magnetic inductor design software are also presented to validate the performance of the developed filter design.

Chapter 4: Results

This chapter presents the results of the variable filter design that is implemented in Chapter 3. These includes the simulation results from Matlab which describe the total harmonic distortions of each design case in the methodology. The core volume, losses and temperature rise results are also presented derived.

Chapter 5: Discussion

This chapter provides an in-depth analysis of the results presented in Chapter 4.

Chapter 6: Conclusion

This chapter concludes the project with a reflection of the work done. It discusses and summarizes the results.

Chapter 7: Recommendation The chapter gives recommendations for additional work to be done on variable filter design. This chapter is followed by the bibliography and Appendix A.

The Matlab simulink model used in this project can be found here: [Kingsley-Macebele-GitHub](#)

Chapter 2

Literature Review

2.1 Introduction

As previously mentioned, the integration of renewable energy sources is increasingly prevalent in modern electrical power systems. These energy sources require interface power converters to transform DC power into AC power suitable for grid integration. However, the use of pulse-width modulation (PWM) techniques in these converters often leads to poor power quality due to the introduction of high-frequency switching harmonics into the grid-injected current [13].

This chapter offers a comprehensive review of various filter topologies and advanced design methodologies aimed at mitigating harmonics in grid-connected inverter systems.

2.2 Filter Topologies for Grid-connected Inverter Systems

Grid-connected inverters necessitate the use of power filters to comply with stringent grid regulations. This requirement arises because these systems are typically controlled by pulse-width modulation (PWM) techniques, which introduce high-frequency harmonics into the grid output current. These harmonics can distort the current, leading to grid oscillations and potential destabilization of the power system [2].

Figure 2.1 illustrates various filter topologies that can be positioned between the inverter and the grid to mitigate output current ripple and reduce harmonics injected into the

2.2. FILTER TOPOLOGIES FOR GRID-CONNECTED INVERTER SYSTEMS

grid. These measures enhance the quality of the grid current at the point of common coupling (PCC). Power filters are generally classified into three categories: passive, active, and hybrid filters. According to the literature [7], active and hybrid filters have been extensively discussed. This section focuses on reviewing basic passive filters—specifically L-filters, LC-filters, and LCL-filters—to identify the optimal filter topology capable of effectively fulfilling the requirements of grid-connected inverter systems.

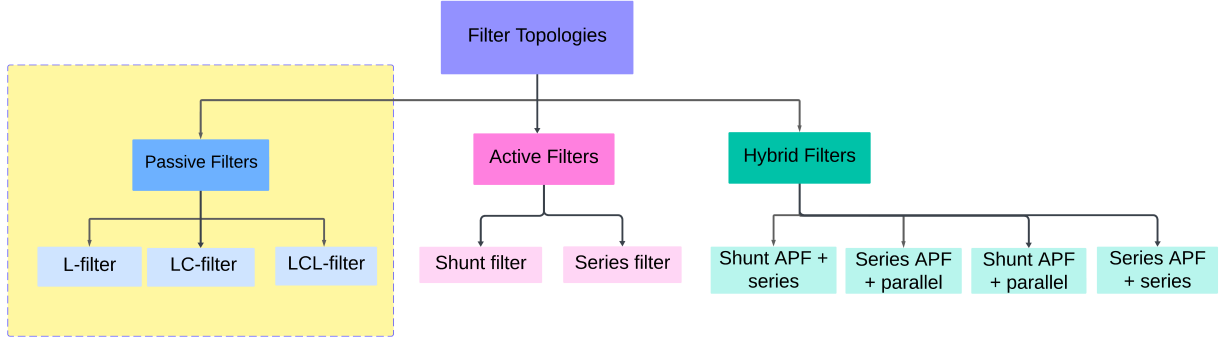


Figure 2.1: Filter topologies for grid-connected inverter systems

2.2.1 L-Filters

L-filters are first-order low-pass filters that play a role in attenuating harmonics of the grid-side current. In the 19th century, L-filters were used as grid interface filters, but later abandoned in most reviewed literature due to several reasons such as poor high-frequency attenuation of -20dB/decade, high voltage drop across the filter, bulky design of the required inductor and their inability to meet the IEEE1547-2008 and IEEE519-1992 regulatory requirements given in [19]-[20] for mid-to-high power inverter applications [2], [6], [18].

In L-filters, the inductor must be large enough to settle the high frequency grid current harmonics, while complying with the TDD requirements ($<5\%$ as per the IEEE-519 standard) [4]. This large inductor results in increased volume and weight of the whole inverter system as well as poor inverter-filter output dynamics [10]. These type of filters can be used in application where there's high switching frequency but sufficient attenuation.

2.2.2 LC-Filters

LC-filters are second-order filters with harmonic attenuation of -40dB/decade or 12dB per octave after the cut-off frequency which makes them more preferable over L-filters. These filters have simple configuration which makes the design process easy [10].

However similar to the L-filters, the LC-filters require a large inductance to achieve a more demanded cut-off frequency of the filter, but with a lower capacitor value. But applying a high capacitor value provide benefits such as improved voltage quality therefore the design of LC filters is a compromise between the inductor and capacitor values. These filters have better damping behaviors. The main disadvantage of this filter is that when its connected to the utility grid, the filter resonance depends on the grid impedance which make this filter not suitable for grid-connected solar inverter systems [10].

2.2.3 LCL-Filters

LCL-filters are third-order filters that effectively smooth out the output current from the voltage source inverter (VSI) [6] and thus, can easily meet the harmonic attenuation requirements for grid-connected inverter systems. These filters are made up of three elements: the inverter-side inductor, grid-side inductor and filter capacitance [4]. Compared to the other two filters mentioned above, LCL-filters offers many advantages that make them preferable over L and LC-filters.

These advantages are: [2], [6], [18].

- Improved efficiency and higher harmonic attenuation of -60dB/decade. This allows lower switching frequencies to be used to meet the grid current constraints [6].
- Their design results in overall size and volume reduction of the total inductance and filter capacitance, which improves the output dynamics of the whole inverter-filter system [5].
- Good performance can be achieved in the range of up to 100 kW of power levels.
- LCL-filters are more economical compared to L and LC-filters.
- Reduced power losses.

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The primary advantage of LCL-filters discussed in most literature is the reduction in the total inductance requirement. Another thing that makes LCL-filters special amongst other filter types are their properties which includes: Rapid dynamic response, low fundamental voltage drop across the filter, high rejection in current ripple and high power factor [2].

With all these benefits, LCL-filters possesses complexities in the selection of its parameters [18]. This is because of the high-order characteristics, various constraints and interrelation of the LCL-filter parameters [2]. Other challenges that arise from LCL-filters are the stability issues resulting from the inherent resonance peak. Therefore, it is necessary to implement damping techniques that will suppress the resonance peak of the LCL-filter.

Damping Methods used in LCL-filters

Due to the LCL-filter resonance frequency, the grid currents are distorted [5]. Different damping techniques are used to successfully remove the resonance frequency. The methods used are the passive damping (PD) methods, active damping methods, filter-based damping methods and state-feedback-based damping methods [2].

The PD method is widely used in many grid-connected inverter applications because it is viewed as the simplest method to implement [6]. However, the main disadvantage of this method is increased power losses because of the use of damping resistors. The three other mentioned methods result in lower losses compared to the PD methods, but at expense of the digital control delay and complexity [1], [2]. In literature, Han et al. [2] thoroughly reviewed and compared various damping methods. But due to the broadness of the damping techniques topic, this section only discusses the passive damping methods.

Passive damping (PD) methods

There are typically six PD damping methods that are used to remove the resonance peak. These methods are implemented by adding a damping resistor in series or parallel into the LCL-filter branches as shown in figure 2.2, [15]. The resulting bode plot before and after the damping methods have been implemented is shown in figure 2.3.

Figure 2.2 shows the damping resistors added in series (R_1 , R_5 , and R_3) and parallel (R_2 , R_6 and R_4) with L_1 , L_2 and C respectively. PD-1 and PD-5 are not usually recommended for grid-connected inverters because of the direct path of power flux through R_1 and R_5 ,

2.2. FILTER TOPOLOGIES FOR GRID-CONNECTED INVERTER SYSTEMS

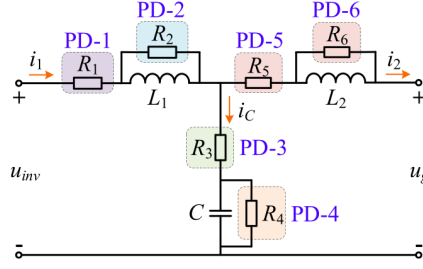


Figure 2.2: Damping resistor placement in six different PD techniques [2]

however it can be noted that R_1 and R_5 can be regarded as the parasitic resistances of the inverter-side and grid-side inductors [2], [8]-[10]. The advantage of PD-1 and PD-5 is the their harmonics attenuation rate does not change for high switching frequencies [2].

PD-2 and PD-6 require high resistances to effectively mitigate the resonance peak [2], [15]. This results in reduced harmonic attenuation ability. Yang et al [2] further discussed fourth passive damping method (PD-4) to be the best in terms of filtering performance. But this method yields high damping losses because of the effect of PCC voltages. Lastly, the third damping method (PD-3) has gained preference in most reviewed literature because it uses a small resistance which results in lower losses, effective damping and better performance regardless of the reduced harmonic attenuation ability. However, the power losses caused by R_3 cannot be avoided. These power losses are as a result of fundamental, switching harmonics and resonance currents [2].

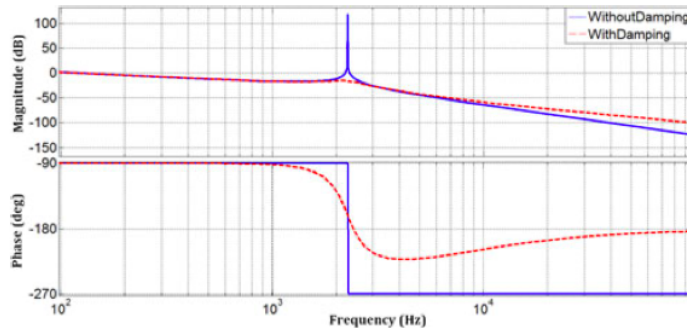


Figure 2.3: Bode plot of LCL-filter with and without damping [1]

Power losses in LCL-filters

The power loss in an LCL filter can be categorized into copper losses and core losses[18]. Core losses primarily arise from high-frequency harmonics in the inductor, allowing for the fundamental frequency to be neglected in core loss calculations [1]. Additionally, core losses are induced by eddy current and hysteresis effects within the magnetic material.

2.3 State-of-the Art LCL-Filter Design Methodologies

This section reviews different state-of-the art LCL-filter designs that have been implemented in literature. Many of these designs follow an iterative process and most often they are customised to operate in different regions and applications[5]. The design proposed by Parikshith & Vinod in 2010 [18] focuses on optimising the power loss and efficiency of the filter. The design provides a standard guideline in choosing the LCL-filter parameters. The inverter-side inductor (L_i) is selected based on the current ripple at switching frequency, and the grid-side inductor (L_g) is selected based on the inverter-side to grid-side current harmonics attenuation while the capacitor is based on the fundamental reactive power limits because the capacitor branch is the dominant path for high frequency harmonics [2]. The researchers also defined the upper and lower limits for the total inductance (L_g) using the fundamental voltage drop across the filter for unity power factor operation and the harmonic current constraints defined by the IEEE519-1992 standard in figure 2.4 respectively . However, this is one of the common approaches of LCL-filter designs as this similar method can be seen in Reznik et.al [6]. In this design, the inverter-side inductance is determined by maximum current ripple which is said to be 10% of the rated current and the grid-side inductance is calculated using grid-side attenuation coefficient (said to be 20%) which will result in 2% of the ripple in the grid-injected current.

Maximum Harmonic Current Distortion in Percent of I_L Individual Harmonic Order (Odd Harmonics)						
I_{sc}/I_L	<11	11<h<17	17<h<23	23<h<35	35<h	TDD
<20*	4.0	2.0	1.5	0.6	0.3	5.0
20<50	7.0	3.5	2.5	1.0	0.5	8.0
50<100	10.0	4.5	4.0	1.5	0.7	12.0
100<1000	12.0	5.5	5.0	2.0	1.0	15.0
>1000	15.0	7.0	6.0	2.5	1.4	20.0

Figure 2.4: Harmonic attenuation limits defined by IEE519-1992 [16]

A similar design is implemented by Han et al. (2019) [2], focusing on the selection of LCL filter parameters, including the total inductance (L_T), current ripple, harmonic attenuation rate (δ), and the reactive power absorbed by the filter capacitor. In their approach, the filter capacitance is selected based on the criterion that the reactive power absorbed by the capacitor is less than 5% of the rated active power of the filter. The capacitor values differ for the two configurations of the LCL-filter: Wye and Delta. Specifically, the capacitance in the Wye configuration is three times that of the Delta configuration.

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In this design, the fundamental voltage drop across the filter ($V_d = I_d(2\pi f_g L_T)$) must be less than 10% of the grid voltage, considering the total inductance (L_T). The inverter-side inductance (L_i) is determined by the inverter topology, modulation factor, and current ripple requirements. Notably, L_i is inversely proportional to the maximum current ripple, necessitating careful selection of the maximum current ripple to optimize performance.

Additionally, the DC-link voltage (V_{dc}) is set high to ensure grid current controllability, particularly when the total inductance is substantial, which can lead to increased switching losses. To meet the design criteria, the resonance frequency (f_{res}) must fall within a specified range, ensuring optimal filter performance and stability.

$$10f_g \leq f_{res} \leq 0.5f_{sw} \quad (2.1)$$

where f_g and f_{sw} are the fundamental grid frequency and switching frequency respectively.

Parameters of the LCL filter	Constraints	Impact of parameters on filter performance
Filter capacitor C_f	$C_f < 5\% \frac{P_n}{2\pi f_g V_{rms}}$	- Large C_f results in low power factor - Small C_f requires large inductance
Total inductance L_T	$L_T \leq 10\% \frac{U_{rms}^2}{2\pi f_g U_{rms}}$	Large L_T results in large voltage drop, small current ripple and PF, poor dynamic response and high cost
Inverter-side inductance L_i	- Single-phase inverter with uni-polar SPWM, $r=8$ - Single-phase inverter with bipolar SPWM, $r=2$	$L_i \geq \frac{U_{dc}}{(20\%-30\%)I_{rated}r_{fsw}}$ $L_i \geq \frac{\sqrt{3}U_{dc}}{12 \times 30\%I_{rated}f_{sw}} m_i$ - Large L_i results in small current ripple, high voltage drop and cost
Harmonic attenuation rate δ $\delta = \frac{ i_{gsw} }{ i_{isw} } = \frac{1}{ 1+a_L(1-L_i C \omega_{sw}^2) }$ $\omega_{sw} = 2\pi f_{sw}$	$\delta = 20\%$	Small δ corresponds to low THD
Resonance frequency f_{res} $\omega_{res} = 2\pi f_{res} = \sqrt{\frac{(L_i + L_g)}{L_i L_g C}}$	$10f_g \leq f_{res} \leq 0.5f_{sw}$	- Small f_{res} results in narrow control bandwidth - Large f_{res} results in resonance peak near f_{sw}

Figure 2.5: Design constraints for the LCL-filter parameter [2]

Another interesting design is the one created by San-Sebastian et al. in 2012 [16], which optimises the volume and cost of the filter using an optimisation function to determine the values of the filter parameters. These values reduces the overall installed power and ensures that filter is compliant with the harmonic attenuation requirements. This design is different from most of the reviewed state-of-the art filter designs because its implemented on a 3.3kV, 8MVA medium-voltage (MV) inverter system. The proposed methodology followed in this design is shown in figure 2.5.

2.3. STATE-OF-THE ART LCL-FILTER DESIGN METHODOLOGIES

Compared to other LCL-filter designs, this design has an advantage in terms of automatically minimising the filter size and cost. However the design process is very complex and another disadvantage is that MV systems require a low switching frequency which excite the resonance frequency very quickly [16].

The cost optimisation functions are as follows,

$$CF1 = L_1 I_{RMS1}^2 + L_2 I_{RMS2}^2 + C_3 V_{RMS3}^2 \quad (2.2)$$

$$CF1 = \frac{i_{L2}}{V_{ab}} VVHD < GridCodeLimit \quad (2.3)$$

Where L_1 , L_2 and C_3 are the inverter-side inductance, grid-side inductance and filter capacitance respectively. I_{RMS1} , I_{RMS2} and I_{RMS3} are the rms values of the current passing through L_1 , L_2 and C_3 respectively. i_{L2} , V_{ab} and $GridCodeLimit$ are the sinusoidal grid-side current, grid phase voltage and harmonic attenuation limits as specified by the IEEE519-1992 standard. $VVHD$ is the Virtual Voltage Harmonic Distortion which indicates the performance of each output voltage harmonic coming from the inverter [16].

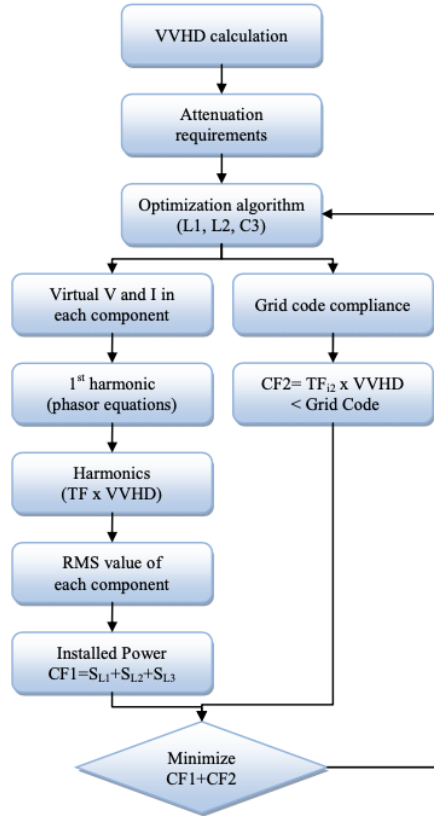


Figure 2.6: LCL filter design algorithm [16]

2.3. STATE-OF-THE ART LCL-FILTER DESIGN METHODOLOGIES

Jayalath & Hanif in 2018 [5] presents a unique design methodology which stipulates that the calculation of the filter's total inductance relies on four critical factors essential for designing an efficient filter

1. Ratio $\mu(\frac{L_g}{L_i})$
2. IEEE-519 standard for harmonic attenuation limits
3. Minimum voltage drop across the filter
4. Reactive power compensation limits

The design methodology is about optimising the total inductance and most importantly the filter capacitance. The results showed the developed methodology to have an advantage over other conventional design methods as shown in figure 2.1.

Table 2.1: Comparison of optimal inductance and capacitance design with conventional design methods

Parameter	Optimum design by Sampath & Hanif in [2018]	Conventional Methods [Design type 1]
Total inductance L_T	Low L_T	High L_T
Filter capacitance, C	Low C	High C
THD [%]	Better THD	Slightly high THD

Magnetic properties for Inductor Design

The selection of LCL-filter parameters alone is insufficient to ensure an efficient filter. Integrating magnetic properties is essential to reduce the volume and cost of the inductors [2], [8]. Magnetic integration techniques involve the careful selection of magnetic materials, core shapes, and winding methods. Table 2.2 compares various soft magnetic materials, highlighting their properties. Among these, powder cores are particularly popular due to their ability to maintain stable inductance under varying temperatures and loads [8]. However, a significant drawback of powder cores is their nonlinear current-dependent characteristics, which can complicate LCL filter design.

Table 2.2: Comparison of magnetic materials [2]

Core Materials	Advantages	Disadvantages
Ferrites	• Low cost • Low core losses • High resistivity	• Fragile • Low saturation flux density • Low curie temperature
Powder cores	• Low eddy currents • High saturation flux density • High resistivity and curie temperature	• Cause fringing effects • Electromagnetic interference
Amorphous alloys	• Low core loss • High saturation flux density	• High cost • Cause fringing effects

2.4 Conclusion

This chapter reviewed various filter topologies utilized in grid-connected solar inverter systems, identifying the LCL-filter as the optimal choice due to its superior harmonic attenuation capabilities and rapid dynamic response, among other benefits. However, several challenges associated with this topology were also highlighted. Additionally, state-of-the-art filter design methodologies were examined, particularly focusing on the approach to selecting filter parameters. Many of these methodologies adopt a similar approach, with fixed design values for the filter inductances. Consequently, the next chapter introduces a developed methodology that varies both the inverter-side and grid-side inductances to determine the optimal design, enhancing the performance and efficiency of the LCL filter.

Chapter 3

Implemented LCL-Filter Design Methodology

As reviewed in the previous chapter, LCL-filters exhibit characteristics that make it possible for them to effectively meet the harmonic attenuation requirements even at lower switching frequencies and provide economical benefits and other advantages such as lower power losses, reduced total inductance and fast dynamic response compared to other filter topologies such as L-filters and LC-filters. However there has been a gap in literature in the development of variable filters incorporating magnetic properties.

The main focus of this project is to develop variable a filter design that test different combinations of the inverter-side L_i and grid-side inductance L_g in terms of the attenuation percentages taken by each inductor to find an optimal design that results in: 1) Reduced total inductor volume, 2) minimum power losses, 3) better temperature rise and lastly, 4) better total harmonic distortion (THD) and improved overall efficiency of the filter. Because of this, the LCL-filter topology was selected for the filter design in this project. The wye-connected configuration is implemented because of its simplicity. Also its design process is documented in many filter designs in literature for grid-connected inverter systems. This chapter discusses the design methodology for the Wye-connected LCL-filter incorporating soft magnetic properties.

3.1 System Description

Firstly, before the design of LCL-filters, it's important to describe the general structure of the grid-connected VSI with Wye-connected LCL-filter shown in figure 3.1. This system consists of L_i and L_g , as the inverter-side and grid-side inductors respectively, C_f is the filter capacitor. In this structure, the PD-3 method was selected because of the advantages mentioned in the previous chapter, to achieve an effective damping of the resonance peak of the LCL-filter. This is represented by the series damping resistor, R_d with C_f . Currents i_i, i_g and i_c are the inverter-side, grid-side and filter capacitor currents respectively. V_i and V_g are the output inverter voltage and grid voltage.

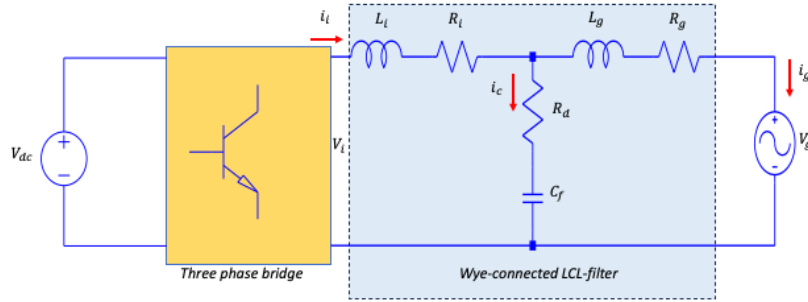


Figure 3.1: LCL-filter Equivalent Circuit of Grid-connected VSI

In this structure, the grid voltage is considered an ideal voltage source at medium and high frequencies, effectively absorbing all harmonics [5], [6]. To determine the worst case scenario of the damping performance of the filter, the parasitic resistances (R_i, R_g and R_d) are set to 0. The resulting transfer function is given below.

$$H_{LCL}(s) = \left| \frac{i_g(s)}{v_i(s)} \right| = \frac{1}{(L_i C_f L_g s^3) + (L_i + L_g)s} \quad (3.1)$$

This transfer function in 3.1 will be used to implement the grid current control for the inverter system. Because PD-3 method is used in this design to remove the resonance peak, the series damping resistor, R_d modifies the transfer function in 3.1 to 3.2.

$$H_{LCL}(s) = \left| \frac{i_g(s)}{v_i(s)} \right| = \frac{C_f R_d + 1}{(L_i C_f L_g s^3) + C_f (L_i + L_g) R_d s^2 + (L_i + L_g)s} \quad (3.2)$$

As shown in table 2.5 in the literature review, the resonance frequency of wye-connected LCL-filters is given in 3.3 and the damping resistor is calculated using the resonance frequency and the filter capacitance as shown 3.4.

$$\omega_{res} = 2\pi f_{res} = \sqrt{\frac{L_i + L_g}{L_i L_g C_f}} \quad (3.3)$$

$$R_d = \frac{1}{3\omega_{res} C_f} \quad (3.4)$$

3.2 Different Design Cases for the variable filter

The filter inductances L_i and L_g were varied based on the harmonic attenuation percentages that each inductor will take to reduce the current ripple from the inverter i.e 50% attenuation by L_i and 50% attenuation by L_g . The design considers 9 different design cases to determine the optimal design combination to improve the filtering performance of the LCL-filter. The performance indexes includes THD, inductor volume, temperature rise, copper and core losses of the filter. The design process will be explained in detail in the following sections. Firstly, the parameter design methodology is followed to determine the fixed total inductance that will satisfy design case 1 resulting in equal attenuation of the current ripple. Then, the variable filter design is implemented to determine varying values of L_i and L_g for the remaining design cases. This is depicted in the algorithm presented in figure 3.5. Lastly, the soft magnetic properties are incorporated in the design to determine the core volume, total losses and temperature rise of the inductors.

Figure 3.2 shows 9 design cases for different attenuation percentages taken by L_i and L_g in reducing the grid-side current ripple. But it is notable that the attenuation percentage of grid-side is taken by both the grid-side inductor L_i and the filter capacitance C_f .

3.2.1 Parameter design Methodology for the Design Cases

The input parameters for the filter design must be known prior the design process. These parameters are, the rated active power P_r , grid-phase voltage V_g , dc-link voltage V_{dc} , grid-operating frequency f_g and switching frequency f_{sw} [2], [6]. The system uses 5kW of rated active power which it makes suitable for low power applications. The other parameters are specified in table 3.1. The base impedance and base capacitance must also be determined in the beginning of the design process as mentioned by Reznik et al. [6].

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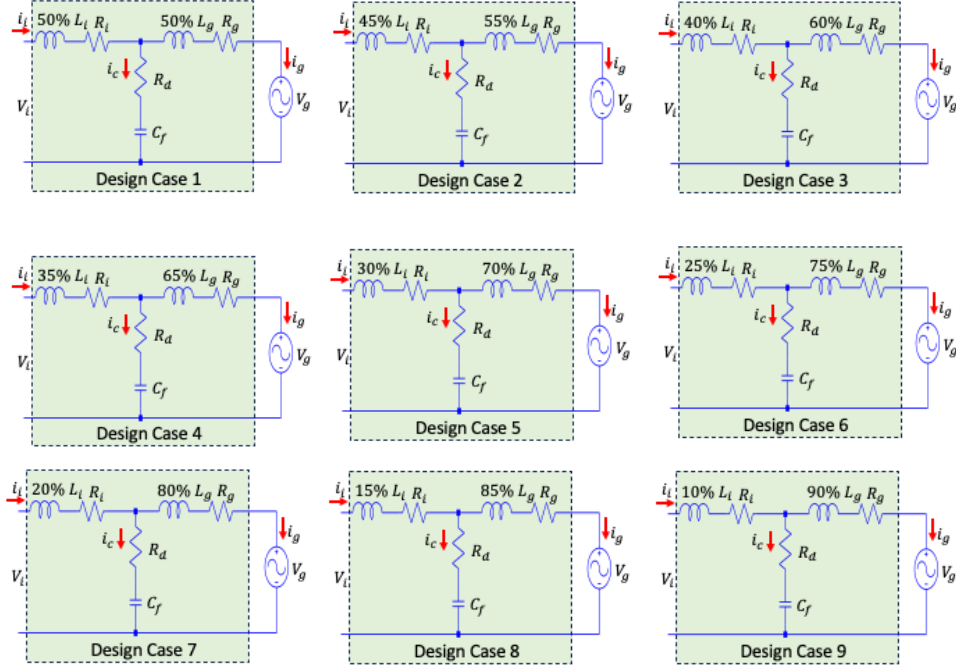


Figure 3.2: Design cases showing different attenuation percentages of L_i and L_g

Calculation of base values

The percentages of the base values are crucial design factors in the designing filter capacitance and the total inductance [6]. The base impedance and the base capacitance of the LCL filter are given by 3.5 and 3.6, where $V_{gl-\ell}$, P_r and ω_g are the line-to-line RMS grid voltage, rated active power and grid-frequency respectively.

$$Z_b = \frac{V_{gl-\ell}^2}{P_r} \quad (3.5)$$

$$C_b = \frac{1}{\omega_g Z_b} \quad (3.6)$$

Using the system specifications in Table 3.1, the values of Z_b and C_b are calculated to be

$$Z_b = \frac{230\sqrt{3}}{5000} = 31.74\Omega$$

$$C_b = \frac{1}{2 \times 50 \times 31.74} = 100.29\mu F$$

In Chapter 1, four key design factors for creating an efficient filter were discussed, one of which is the ratio μ . A primary goal of this project is to minimize the inductor volume. Therefore, a ratio of $\mu = 1$ was selected, as recommended by [1] and [5]. This choice

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ensures minimal total inductance and filter capacitance, reduced voltage drop, decreased switching losses, minimized reactive power production, minimal energy storage in the LCL filter, and improved dynamic response to variations in grid impedance. These objectives are achieved without compromising the harmonic attenuation specified by the IEEE-519 standard [1].

Ratio $\mu = 1$ results in the minimum product of $L_T C$ given by the formula in equation 3.7. Where k is the the ratio of the switching frequency to the resonance frequency ($\frac{f_{sw}}{f_{res}}$) used to determine the range in which the resonance frequency lies. Usually, the value of k is within the range $1 < k < 9.5$ for double update PWM inverter [1].

$$L_T C = \frac{k^2}{4\pi^2 f_{sw}^2} \frac{(1 + \mu)^2}{\mu} \quad (3.7)$$

The minimum total inductance is also shown in 3.8.

$$\ell_{Tmin} = \frac{|v_{i(pu)}(h)|}{\rho |i_{g(pu)}(h)| |1 - k^2|} \quad (3.8)$$

Where $v_{i(pu)}(h)$ is the switching ripple component in $p.u$ and its given as $v_{i(pu)}(h) = \frac{v_i(h)}{V_g}$, where $V_i(h)$ can be approximated as $V_{dc}/4$, ρ is the ratio of the switching frequency and fundamental grid-frequency or base frequency ($\frac{f_{sw}}{f_g}$) and $i_{g(pu)}(h)$ is grid-current harmonics attenuation limits specified by IEEE519-1992 standard.

From the minimum inductance, the typical value of the total inductance can be found. This total inductance must be greater than the minimum inductance ($L_T \geq L_{Tmin}$).

$$L_T = \frac{\ell_T Z_b}{2\pi f_g} \quad (3.9)$$

The filter capacitance plays an important in minimising the reactive power of the inverter and also improving the power factor (PF) such that its close to 1(unity) [1]. Therefore the value of C_f is selected to limit the overrating of the inverter and for reactive power compensation. The allowed power factor variation is usually given as 5%, therefore the filter capacitance, therefore the value of capacitance must satisfy the following range to prevent a decrease in the PF.

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$$0 \leq C_f \leq 0.05C_b \quad (3.10)$$

Figure 3.3 shows the design methodology used to determine the total inductances for design case 1 whereby both the inverter-side inductance L_i and the grid-side inductance L_g have equal attenuation percentage of 50%. The following steps are followed to determine the values of L_i and L_g for 50% equal attenuation.

1. First determine the input system parameters as mentioned in the beginning of this chapter, also shown in table 3.1.
2. Determine the damping method, ratio μ , control method and specify the IEEE519 harmonic limits as required by your design.
3. First determine the input system parameters as mentioned in the beginning of this chapter, also shown in figure 3.3.
4. Calculate the base impedance and base capacitance as described previously.
5. Find an estimate of the k value for the resonance frequency lies which must satisfy the range: $1 < k < 9.5$. Then calculate the value of $L_T C$ using equation 3.7 to meet the k value.
6. Calculate the minimum value of the total inductance ℓ_{Tmin} using equation 3.8.
7. Use the value of $L_T C$ to find the maximum capacitance of the filter corresponding to the minimum inductance
8. Set the reactive power limit and then determine the final value of the capacitor and total inductance.

Table 3.1: Input system parameters

Input parameter type	Value
Rated active power (P_r)	5 kW
Grid phase voltage (V_g)	230 V
DC link voltage (V_{dc})	480 V
Grid frequency (f_g)	50 Hz
Grid switching frequency (f_{sw})	10 kHz

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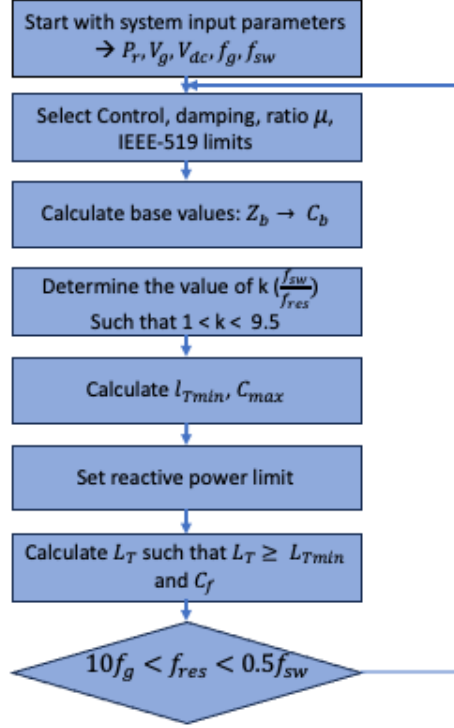


Figure 3.3: Design flow to determine the total Inductance $L_T = L_i + L_g$ and filter capacitance C_f parameters for design case 1 for which both L_i and L_g have 50% attenuation of the current ripple

The parameter design steps were followed according to the algorithm in Figure 3.3 for design case 1.

1. The grid-side current control is selected.
2. As mentioned earlier, the PD-3 method is used because of its controller simplicity and effectiveness in resonance damping while using a smaller resistance.
3. Since the designed LCL-filter operate as a single filtering unit, ratio $\mu = 1$ is used based on the advantages that were reviewed earlier.
4. Harmonic attenuation limits are based on the IEEE-519 standards shown in Table 3.2. The inverter is designed to reject harmonic currents that exceed 0.3% of the rated injected grid current above the 50th harmonics ($< 2500Hz$).

Table 3.2: IEEE519 Harmonic attenuation limits

Limits as a percentage of rated current amplitude						
$\frac{I_{sc}}{I_L}$	$3 \leq h \leq 11$	$11 \leq h \leq 17$	$17 \leq h \leq 23$	$23 \leq h \leq 35$	$35 \leq h \leq 50$	TDD
$< 20^*$	4.0	2.0	1.5	0.6	0.3	5.0

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5. The base values are calculated earlier using equation 3.5 and 3.6 and were found to be $Z_b = 31.74\Omega$ and $C_b = 100.29\mu F$
6. The value of k must bring about a balance in the power losses in the passive damping scheme, minimum stored energy, etc [1]. Based on the harmonic constraints for the inverter discussed in step 4, initially the resonance frequency is chosen to be 2500 Hz. In this design, k is calculated to be $k = \frac{f_{sw}}{f_{res}} = \frac{10000}{2500} = 4$. This value satisfies the condition $1 < k < 9.5$.
7. The product of $L_T C$ is now calculated using equation 3.7 by substituting the value of k . Therefore, $L_T C = \frac{4^2}{4\pi^2 10000^2} = 4.05 \times 10^{-9}$.
8. Using equation 3.8, the minimum total inductance per unit needed to meet the specified harmonic attenuation requirements by IEEE-519 is calculated as follows.

$$v_{i(pu)}(h) = \frac{v_i(h)}{V_g}, \text{ where } V_i(h) = V_{dc}/4 = \frac{480}{4} = 120 \text{ and } V_g = 230 \text{ V, thus } v_{i(pu)}(h) = 0.52, \rho = \frac{f_{sw}}{f_g} = \frac{10000}{50} = 200 \text{ and } i_{g(pu)}(h) \text{ is grid-current harmonics attenuation limits specified by IEEE519-1992 standard. In this case, } i_{g(pu)}(h) = 0.3\%$$

Therefore, the minimum inductance is given as

$$\ell_{Tmin} = \frac{|0.52|}{200|0.003||1-4^2|} = 0.0577mH.$$
9. From the calculated product of $L_T C$, the maximum filter capacitance for the harmonic limitations is found to be $C_{max} = 69.828\mu F$.
10. However for the 5% maximum power factor variations as seen by the grid. The filter capacitance need to be within the 5% limit of the base capacitance C_b . Therefore, $C_f = 0.05C_b = 5.015\mu F$. In order to compensate for the inductive reactance of the LCL-filter, a higher power factor variation can be used [6].
11. The resulting L_T is now found to be $L_T = \frac{\ell_T Z_b}{2\pi f_g} = 5.87983mH$ for the corresponding filter capacitance $C_f \approx 5\mu F$. Therefore, $L_i = 2.928mH$ and $L_g = 2.928mH$.

Both $L_i = 2.928mH$ and $L_g = 2.928mH$ will have 50% current ripple attenuation rejection. These values will then be used as the base to design L_i and L_g for the remaining 8 design cases by varying them using the methodology described in the following section.

3.2.2 Variable LCL-filter design based on different attenuation percentages

Reznik et al [6] methodology was modified so that the inverter-side and grid-side inductances ($L_i = 2.928mH$ and $L_g = 2.928mH$) calculated in the previous section can be varied to

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get different attenuation percentages as indicated by the design cases in figure 3.2. In order to vary the inverter-side inductance L_i and grid-side inductance L_g , the ripple of the rated current $I_{r(ripple)}$ and the desired attenuation coefficient k_a are varied respectively as shown in figure 3.4 and the product of these two values must equal to 0.3% of the rated injected grid current as specified by the IEEE-519 standard. The maximum rated current is given as

$$I_{r(max)} = \frac{P_r \sqrt{2}}{3V_g} \quad (3.11)$$

Where P_r is the rated active power and V_g is the grid-phase voltage. Therefore, $I_{r(max)} = \frac{5000\sqrt{2}}{3 \times 230} = 10.25A$.

Equation 3.12 shows the maximum current ripple for dc/ac inverter. This equation can be further simplified by noticing that the maximum peak-to-peak current ripple occurs at $m=0.5$, the modulation factor for typical SPWM inverters [6].

$$\Delta I_{r(ripple)max} = \frac{2V_{dc}}{3L_i}(1-m)mT_{sw} \quad (3.12)$$

$$\Delta I_{r(ripple)max} = \frac{V_{dc}}{6f_{sw}L_i} \quad (3.13)$$

From equation 3.13, the current ripple $I_{r(ripple)}$ can now be changed to vary L_i and by rearranging this equation, L_i is given as

$$L_i = \frac{V_{dc}}{6f_{sw}\Delta I_{r(ripple)max}} \quad (3.14)$$

where $\Delta I_{r(ripple)max} = I_{r(ripple)}I_{r(max)}$

In order to vary L_g , the attenuation of high-frequency switching harmonics is considered. This is deduced by first considering the inverter as a harmonic generator or a current source and the grid as a short-circuit for every harmonic frequency [1],[6]. The grid-side current attenuation coefficient k_a is then given as the ratio of the grid-side current harmonics and the inverter-side current harmonics. The grid-side inductance can now be calculated using equation 3.16.

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$$\frac{i_g(h)}{i_i(h)} = \frac{1}{|1 + r[1 - L_i C_b \omega_{sw}^2 x]|} = k_a \quad (3.15)$$

where x is maximum power factor variations of the grid. In our design, $x = 5\%$.

$$L_g = \frac{\sqrt{\frac{1}{k_a^2} + 1}}{C_f \omega_{sw}^2} \quad (3.16)$$

In equation 3.16, the relationship between the grid-side current attenuation coefficient k_a and the grid-side inductance L_g is inversely proportional ($L_g \propto \frac{1}{k_a}$). Therefore k_a is increased by 5% so that the grid-side inductance drops. Because the grid-side inductance need to be less than the inverter-side inductance to ensure stability in the system [18]. The inverter-side inductance is increased by reducing the current ripple $I_{r(ripple)}$ by 5% as shown by the algorithm in Figure 3.5.

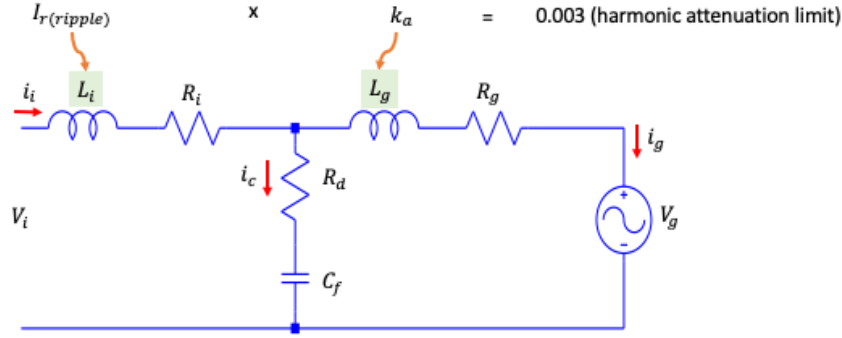


Figure 3.4: Illustration of the variable filter design using $I_{r(ripple)}$ and k_a to vary L_i and L_g respectively.

The design steps followed to vary L_i and L_g are described below.

1. Start with the inverter-side inductance L_i and L_g and filter capacitance C_f found using the design procedure in Figure 3.3. These values are for design case 1 where both L_i and L_g have 50% current ripple attenuation rejection.
2. Using equation 3.14, use the value of L_i and calculated rated current to determine the current ripple $I_{r(ripple)}$.
3. To determine the grid-current attenuation coefficient k_a , use the value of L_g and filter capacitance C_f in equation 3.16.
4. The calculated values of $I_{r(ripple)}$ and k_a are now varied, whereby $I_{r(ripple)}$ is reduced by 5% to increase the inverter-side inductance and k_a is increased by 5% to decrease

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the value of L_g . The product of two values must always equal to 0.3% as specified by the IEEE519 standard.

5. The new values of $I_{r(ripple)}$ and k_a are now substituted back in equation 3.14 and equation 3.16 to find L_i and L_g respectively.
6. The previous step is repeated 8 times for the other design cases.
7. For each step, determine the resonance frequency to make sure that its equal to the initial resonance frequency from the initial values of L_i and L_g
8. If not equal, change the value of the filter capacitance, C_f to get the same resonating point. The value of the capacitor must be in the range: $0 < C_f < C_{max}$ as calculated in the previous design.
9. Finally, validate the design using the resonance frequency range: $10f_g \leq 0.5f_{sw}$
10. Determine the passive damping resistor for each design step using equation 3.4

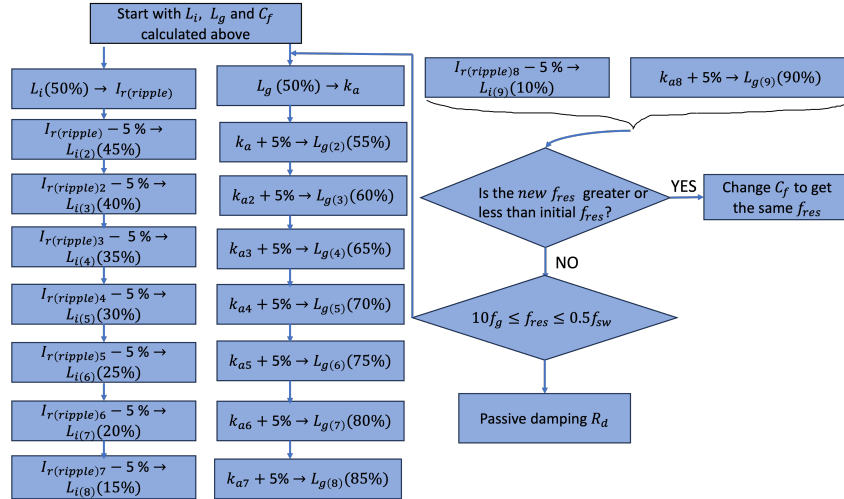


Figure 3.5: Design flow to determine variable inverter-side inductance L_i and grid-side inductance L_g for different attenuation percentages

The design procedure is followed below for the design cases using the design flow in Figure 3.5.

1. The values of $L_i = 2.928mH$, $L_g = 2.928mH$ and $C_f = 5.01\mu F$ calculated in the design procedure described previously are used where each initially has 50% attenuation percentages.
2. Using $L_i = 2.928mH$ and $I_{r(max)} = 10.25$ A, the current ripple $I_{r(ripple)}$ is calculated as follows
$$I_{r(ripple)} = \frac{480}{6 \times 10000 \times 10.25 \times 2.928 \times 10^{-3}} = 0.266$$

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3. Using $L_g = 2.928mH$ and $C_f = 5.01\mu F$, the grid-side current attenuation coefficient k_a is calculated as follows using equation 3.16. The value of $k_a = 0.018$. The product of $I_{r(ripple)}$ and $k_a = 0.3\%$
4. Now vary the values of $I_{r(ripple)}$ and k_a by decreasing and increasing them by 5%. $I_{r(ripple)} - 5\% = 0.2527$ and $k_a + 5\% = 0.0198$. These values results in 45% and 55% attenuation percentages for L_i and L_g as shown in design case 2, figure 3.2.
5. The values of $I_{r(ripple)} = 0.2527$ and $k_a = 0.0198$ are now substituted back into equation 3.14 and 3.16. This results in $L_i = 3.1mH$ and $L_g = 2.5mH$.
6. The resonance frequency of $L_i = 2.928mH$ and $L_g = 2.928mH$ with $C_f = 5.01\mu F$ for design case 1 is found to be 1866.4 Hz using equation 3.3.
7. To get the same resonance frequency of 1866.4 Hz, the filter capacitance C_f is varied in the range: $0 < C_f < 69.828\mu F$ as calculated in the previous method for design case 1. The filter capacitance is found to be $C_f = 5.32\mu F$.
8. The series damping resistor is $R_d = R_f = 5.34\Omega$ using equation 3.4 for design case 2.
9. The steps outlined can be repeated to find the values of L_i and L_g for the remaining cases. However, this is a very tedious process, so a Matlab script in *Appendix A* is created to calculate the values. Table 3.3 shows the current ripples and grid-side current attenuation coefficients used for all the design cases while table 3.4 shows the calculated values of L_i and L_g for the design cases using the design procedure described in above. The filter capacitance C_f , damping resistor R_d and the resonance frequency f_{res} are also shown in Table 3.5.

Table 3.3: Values of the current ripple $I_{r(ripple)}$ and k_a used to determine L_i and L_g for the design cases

Design cases	Current ripple $I_{r(ripple)}$	Grid-side current attenuation coefficient k_a
1	26.6%	1.80%
2	25.27%	1.98%
3	23.94%	2.16%
4	22.61%	2.34%
5	21.28%	2.52%
6	19.95%	2.70%
7	18.62%	2.88%
8	17.29%	3.06%
9	15.96%	3.24%

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Table 3.4: Variable filter design inverter-side L_i and grid-side inductor values L_g for the design cases

Design cases	Inverter-side inductance (L_i) [mH]	Grid-side inductance (L_g) [mH]	$r\left(\frac{L_g}{L_i}\right)$
1	2.9	2.9	0.97
2	3.1	2.5	0.79
3	3.3	2.1	0.65
4	3.5	1.8	0.52
5	3.7	1.5	0.42
6	3.9	1.3	0.33
7	4.2	1.0	0.24
8	4.5	0.78	0.17
9	4.9	0.54	0.11

Table 3.5: The filter capacitance C_f , damping resistor R_d and resonance frequency f_{res} for the design cases

Design cases	Filter capacitance C_f [μF]	Damping Resistor $R_d = R_f$ [Ω]	Resonance frequency (f_{res}) [Hz]
1	5.01	5.66	1868.1
2	5.32	5.34	1866.4
3	5.68	5.00	1866.4
4	6.13	4.64	1866.4
5	6.74	4.22	1866.4
6	7.57	3.75	1866.5
7	8.83	3.22	1866.4
8	10.91	2.61	1866.4
9	15.01	1.89	1866.4

3.2.3 Developed MATLAB Simulink model to analyse the performance of filter

A Matlab Simulink Model was developed to analyse and validate the performance the design cases discussed above in terms of the attenuation requirements of the system which has to be $< 5\%$ as per the IEEE519-1992 Standard. The model is shown in figure 3.7. In the model, L_1 represent the inverter-side inductance and L_2 represent the grid-side inductance and C_f is the filter capacitance. The parameters values are calculated using the Matlab script in Appendix A. The powergui is used to perform FFT analysis to get the THD performance of each design case.

The proportional-integral (PI) Controller is used for the grid-injected current regulation [13]. The transfer function in equation 3.2 with damping is used to find the implement

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the PID tuning of the current loop. Because the implemented system is grid-connected, the controller is synchronised with the voltage of the grid using phase-locked loop (PLL) in the blue-coloured block in figure 3.7. The block diagram is clearly shown in Appendix A.

```
% Current loop LCL with damping resistance
Gi = (s*Cf1*Rf+1)/(s^3*L2*L1*Cf1+s^2*Rf*(L1+L2)*Cf1+s*(L2+L1));
41
% input is Uinv and output is current of L2
%% Current control tuning using PI controller (The current is measured at
    (LCL) L2)
C = pidstd(1,1);
C = pidtune(Gi,C,100*2*pi);
Gi_op = C*Gi; % open loop
Gi_fb = feedback(C*Gi,1); % closed loop
%% Using sisotool
Gpi = 29.662*((1+0.022*s)/s); % PI tuned using sisotool
Gi_op_sstool = Gpi*Gi;
Gi_fb_sstool = feedback(Gi_op_sstool,1);
```

The resulting controller gains, are $K_p = 0.3$ and $K_i = 40$

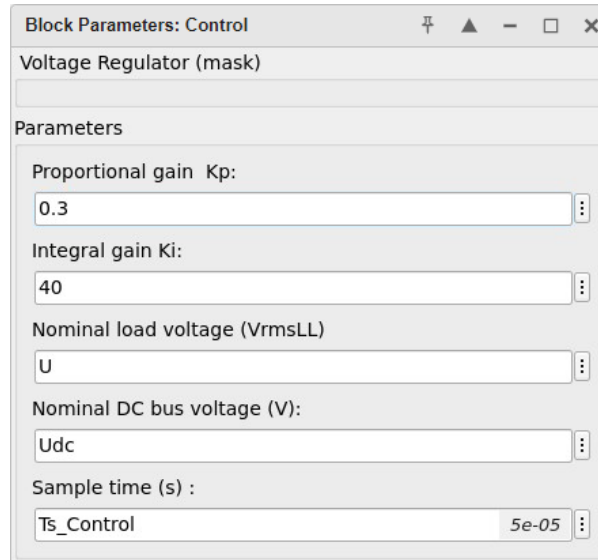


Figure 3.6: Controller gains for the Matlab Simulink Model

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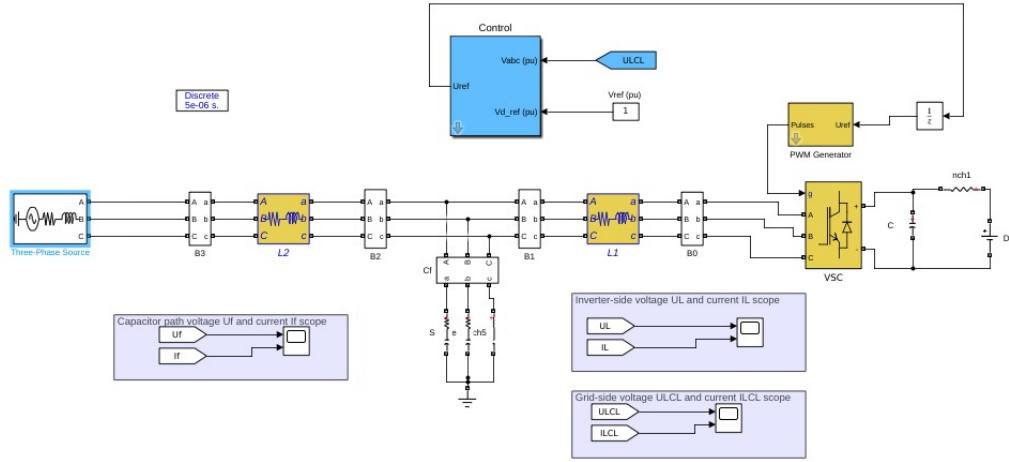


Figure 3.7: Matlab simulink model of the grid-connected inverter system

3.2.4 Magnetic Integration for the LCL-filter design

As discussed in the literature review in chapter 1, magnetic integration in inductors is mainly focused on the selection of magnetic materials, core shapes, etc. The main reason of integrating magnetic materials in this design is to determine the optimal design case that result in minimum inductor volume and power losses while achieving stable inductance against temperature variations. Thus, high-flux core type which is a soft powder core material was used to achieve low eddy current and hysteresis losses.

The guidelines for designing inductors are specified by manufacturers [5]. A magnetics inductor design tool was used to determine the inductor volume, losses and temperature rise with selected optimum powder cores for the different design cases.

Part No	Specification	Core OD(mm)	Size Code
58617	Datasheet	62	620
58616	Datasheet	62	620
58672	Datasheet	77.8	866
58668	Datasheet	77.8	866
58612	Datasheet	77.8	906
58608	Datasheet	77.8	906

Selected Part Numbers			
Part No	58617	Core OD(mm)	62
OD	62.91	mm	Perm
ID	31.69	mm	AL
HT	25.91	mm	μH/T²

Design Output	
Min Inductance @ Full Load (μH):	2552.5
Nom Inductance @ No Load (μH):	4838.4
Min Inductance @ Specified Current (μH):	2855.5
Core Loss (W):	0.745
Copper Loss (W):	14.587
Total Losses (W):	15.332
Temperature Rise (°C):	55.7
Number of Turns:	160
Wire Size (AWG):	14
Winding Factor (%):	46.8%
DC Resistance (mΩ):	136.83
Finished OD (mm):	78
Finished HT (mm):	41
Total Wire Length (mm):	16,523

Figure 3.8: Magnetics inductor design tool

3.2. DIFFERENT DESIGN CASES FOR THE VARIABLE FILTER

The design tool is divided into the three sections. The first section is for the design input parameters (rated current, switching frequency, and inductance). The second section is for different core size for the chosen core material. After input the design parameters, the inductor volume, losses and temperature rise of the inductor are determined in the third section (referred to as Design output).

Chapter 4

Results

To validate the values from table 3.5, the bode plot of the LCL-filter in Figure 4.1 was plotted in Matlab to show the undamped system performance. In this figure, the resonance peak is at the same point (1866.4 Hz) for all the design cases. The harmonic current ripples are attenuated at the same rate.

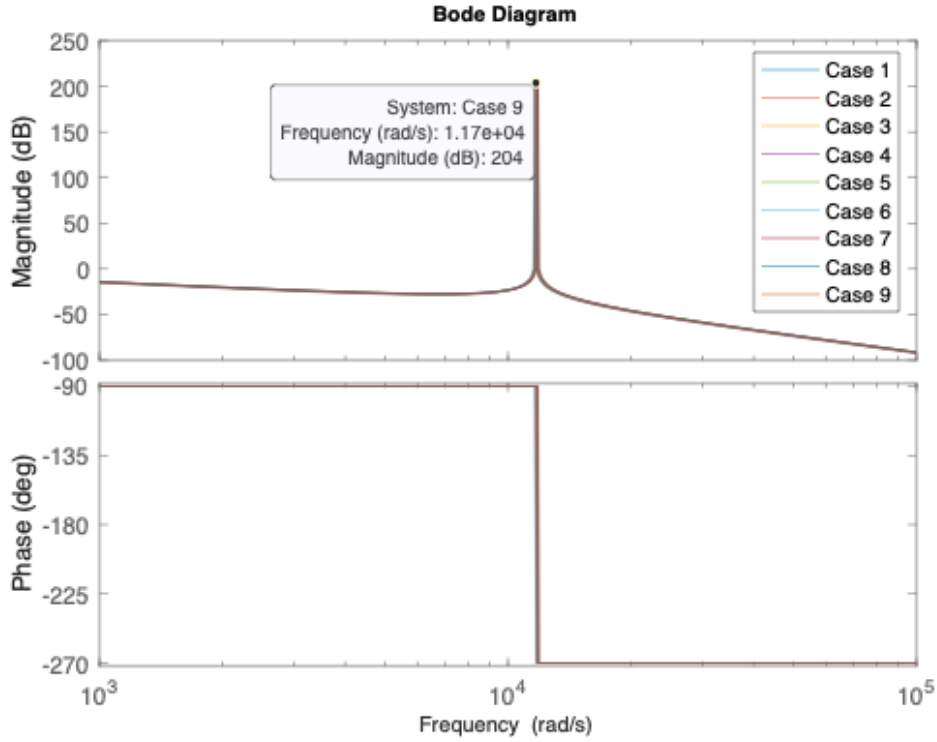


Figure 4.1: Bode plot of the LCL-filter without passive damping for the design cases

In Figure 4.2, the damping resistor $R_d = R_f$ have been inserted in series with the filter

capacitor and it varies from $R_f = 5.66\Omega$ to $R_f = 1.89\Omega$ from case 1 to case 9. The results shows that the high gain spikes at the resonance frequency has been effectively suppressed by all these damping resistors. Resulting in the smoothing of the overall phase response. As the filter capacitance was increased, the damping resistor R_f decreases from design case 1 to design case 2 to support the inversely proportional relationship in equation 3.4.

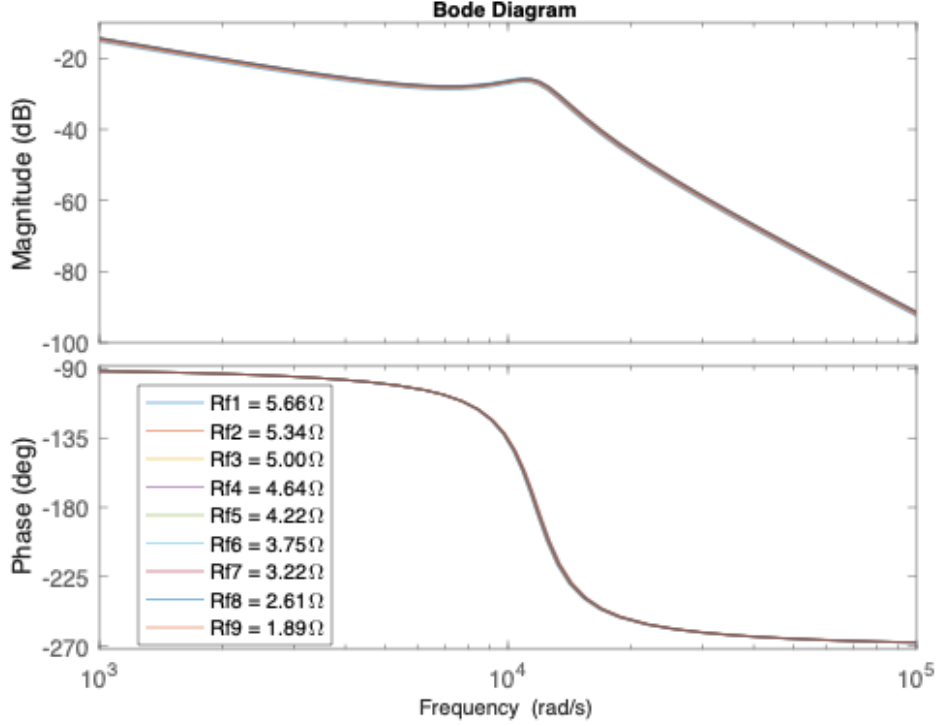


Figure 4.2: Bode plot of the LCL-filter with passive damping for the design cases

4.1 Simulation Results

Design Case 1

The design cases were tested and validated on a 5kW grid-connected inverter system in Matlab simulink. The inductors used in design case 1 are $L_i = 2.9mH$ and $L_g = 2.9mH$ with equal attenuation percentage of 50%. Figure 4.3 shows their corresponding THD plots. It is clearly depicted that the grid-side high-frequency components are below 0.3% as specified by IEEE519. The THD is 3.35% which agrees with the 5% THD requirements limits.

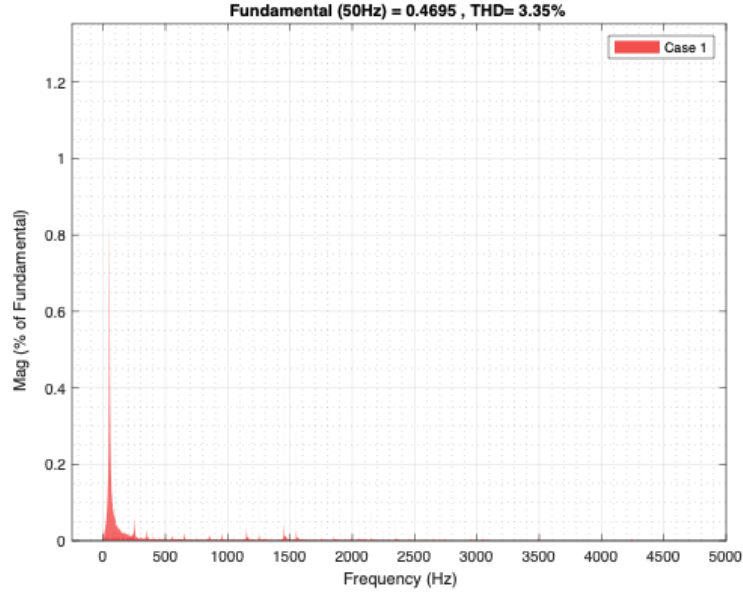


Figure 4.3: THD plot for design case 1

Design Case 2

The inductors used in design case 2 are $L_i = 3.1mH$ and $L_g = 2.5mH$ with attenuation percentages of 45% and 55% respectively. Figure 4.4 shows a THD of 3.25%.

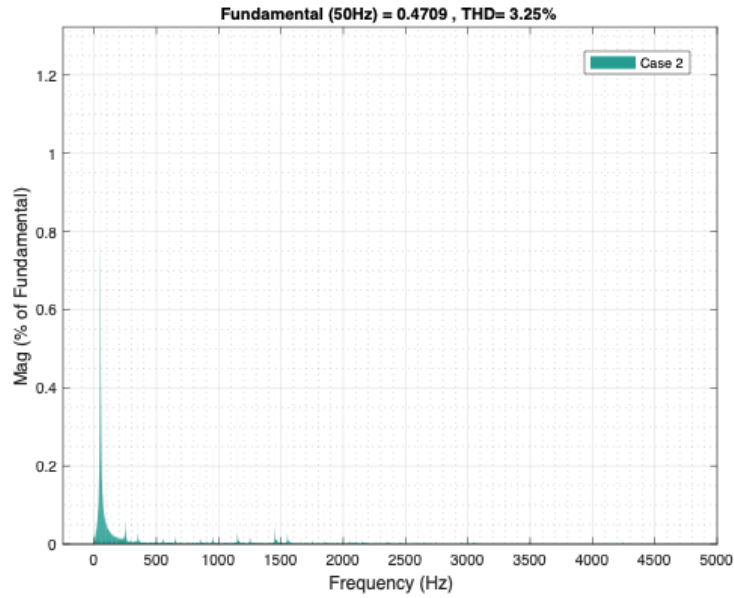


Figure 4.4: THD plot for design case 2

Design Case 3

The inductors used in design case 3 are $L_i = 3.3mH$ and $L_g = 2.1mH$ with attenuation percentages of 40% and 60% respectively. Figure 4.5 shows a THD of 3.23%

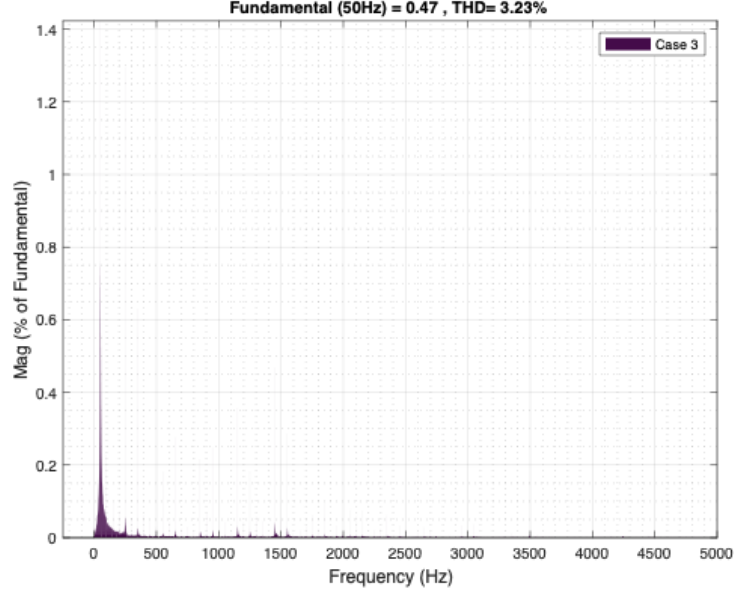


Figure 4.5: THD plot for design case 3

Design Case 4

The inductors used in design case 4 are $L_i = 3.5mH$ and $L_g = 1.8mH$ with attenuation percentages of 35% and 65% respectively. Figure 4.8 shows a THD of 3.22%. The inverter-side and grid-side currents are shown in figure 4.6 and 4.7. Both meet the harmonic attenuation limits defined by IEEE519 of THD less than 5%.

Design Case 5

The inductors used in design case 5 are $L_i = 3.7mH$ and $L_g = 1.5mH$ with attenuation percentages of 30% and 70% respectively. Figure 4.9 shows a THD of 3.16%.

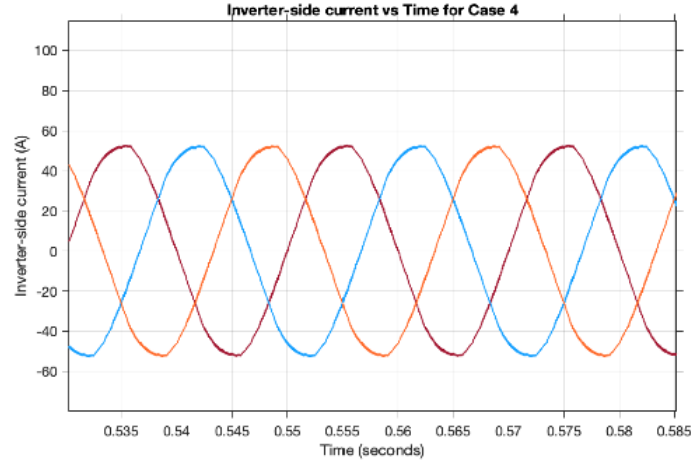


Figure 4.6: Inverter-side current for design case 4

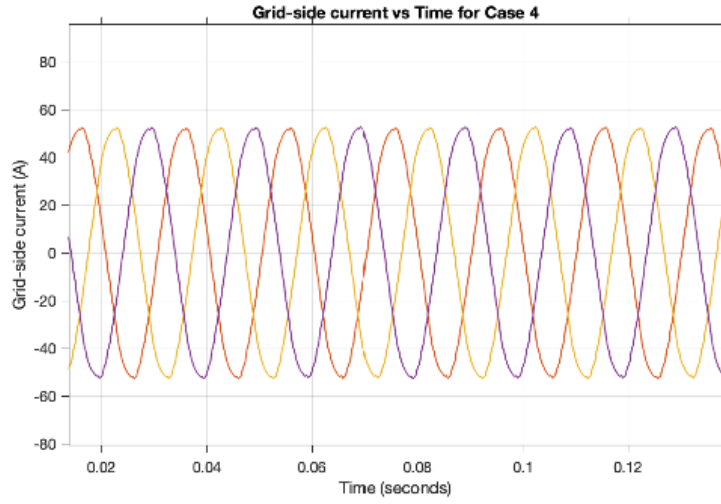


Figure 4.7: Grid-side current for design case 4

Design Case 6

The inductors used in design case 6 are $L_i = 3.9mH$ and $L_g = 1.3mH$ with attenuation percentages of 25% and 75% respectively. Figure 4.10 shows a THD of 3.18%.

Design Case 7

The inductors used in design case 7 are $L_i = 4.2mH$ and $L_g = 1.0mH$ with attenuation percentages of 20% and 80% respectively. Figure 4.11 shows a THD of 3.18%.

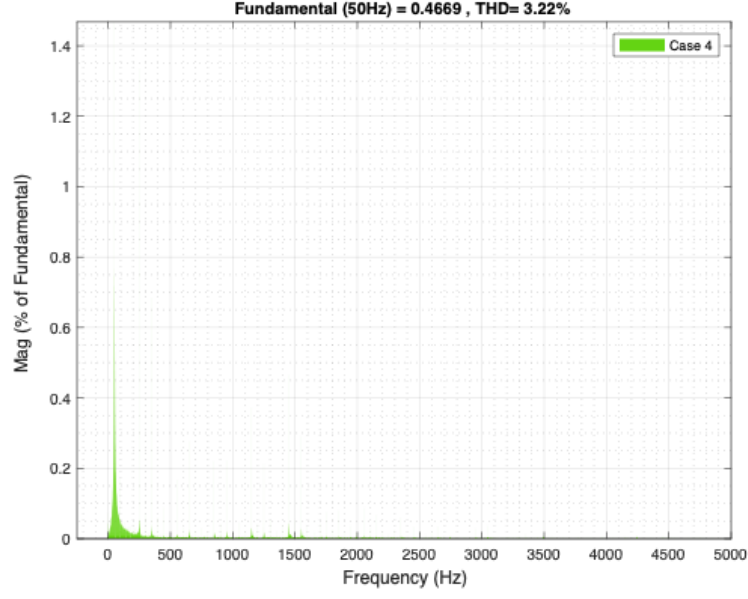


Figure 4.8: THD for design case 4

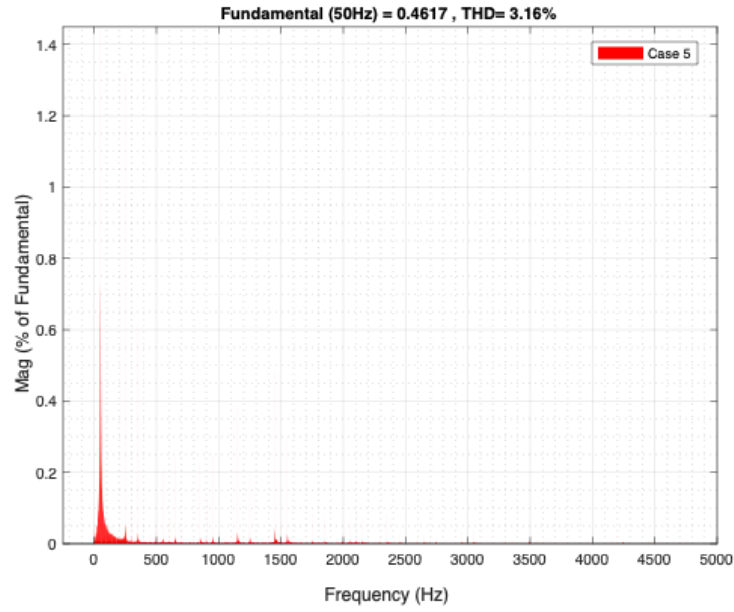


Figure 4.9: THD plot for design case 5

Design Case 8

The inductors used in design case 8 are $L_i = 4.5mH$ and $L_g = 0.78mH$ with attenuation percentages of 15% and 85% respectively. Figure 4.12 shows a THD of 3.17%.

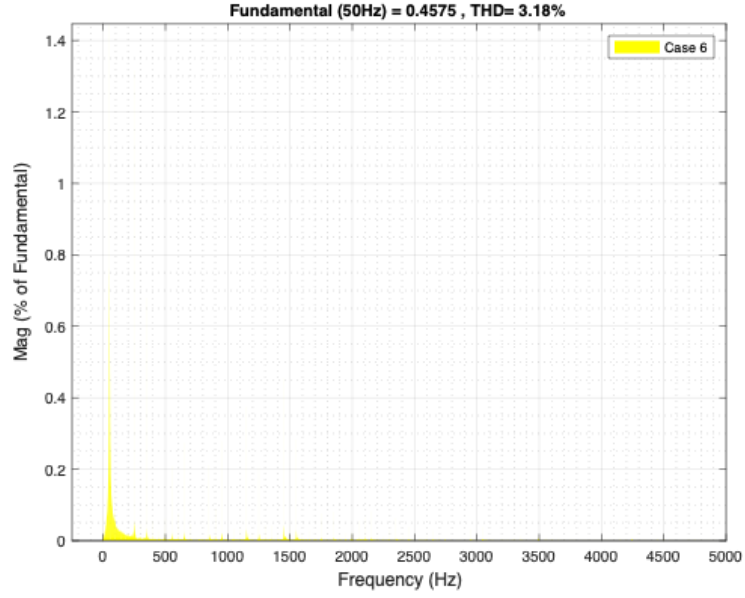


Figure 4.10: THD plot for design case 6

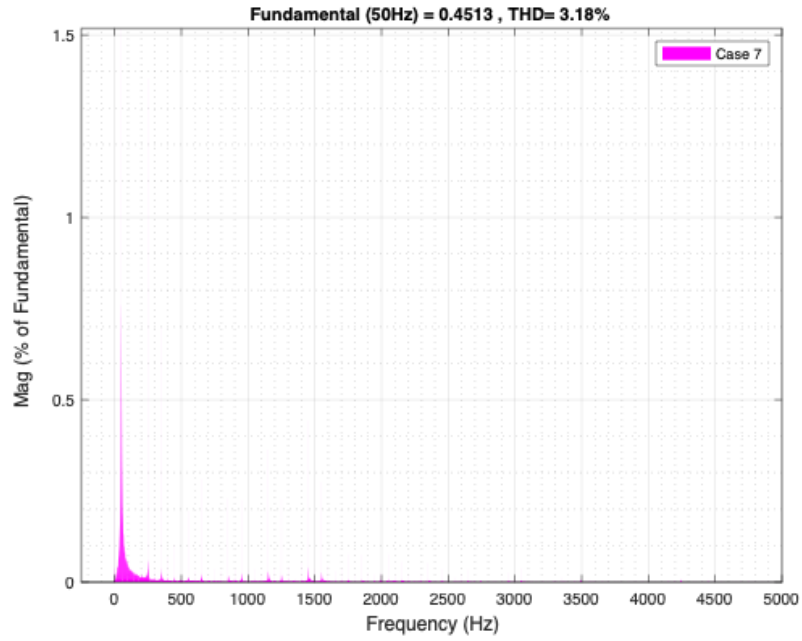


Figure 4.11: THD plot for design case 7

Design Case 9

The inductors used in design case 9 are $L_i = 4.9mH$ and $L_g = 0.54mH$ with attenuation percentages of 10% and 90% respectively. Figure 4.13 shows a THD of 3.20%.

4.2. DESIGN CASES INCORPORATING POWDER CORE MAGNETIC PROPERTIES

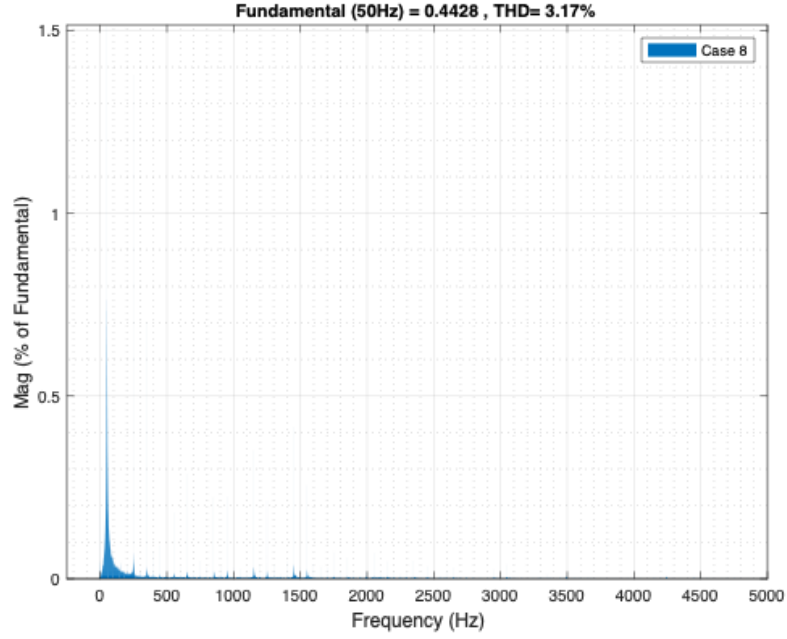


Figure 4.12: THD plot for design case 8

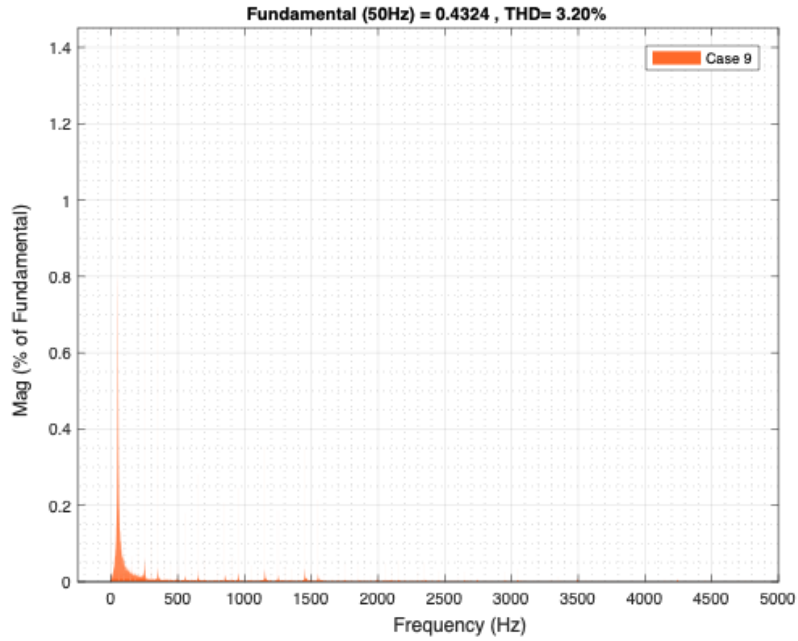


Figure 4.13: THD plot for design case 9

4.2 Design cases incorporating powder core magnetic properties

Figure ?? shows the results found using the magnetics inductor design software. The design cases are compared using the high flux powder cores. The wire size AWG 12 is

4.2. DESIGN CASES INCORPORATING POWDER CORE MAGNETIC PROPERTIES

used to handle the rated grid-side current. The design cases are ranked from 1-9 for each performance index, where 1 indicates the best design case combination for that index. In these results, it can be noticed that there are trade-offs between power loss optimisation, temperature rise and core volume.

Table 4.1: Comparison of the design cases in terms of core volume, losses and temperature rise

Design cases	Core Type		Core Volume [mm^3]		Copper Loss [W]		Core Loss [W]		Total Loss [W] (<i>copper</i> + <i>core</i>)		Temperature Rise [$^{\circ}C$]	
	L_i	L_g	L_i	L_g	L_i	L_g	L_i	L_g	L_i	L_g	L_i	L_g
1	58617	58617	60039.27	60039.27	15.031	15.031	0.755	0.755	15.781	15.781	57.1	57.1
2	58617	58617	60039.27	60039.27	15.895	12.796	0.690	0.621	16.585	13.417	59.4	49.8
3	58617	58617	60039.27	60039.27	16.758	11.077	0.625	0.498	17.383	11.576	61.8	44.1
4	58617	58599	60039.27	41593.81	16.736	10.820	0.562	0.335	17.299	12.155	61.6	54.5
5	58617	58599	60039.27	41593.81	16.716	11.846	0.500	0.266	17.217	10.308	61.3	47.5
6	58617	58110	60039.27	25298.01	16.697	10.042	0.440	0.168	17.137	12.014	61.1	67.3
7	58617	58716	60039.27	19398.36	16.680	10.089	0.379	0.111	17.059	10.200	60.9	68.2
8	58775	58172	100530.03	437763.901	23.856	10.304	0.472	0.111	24.328	10.415	56.9	11.8
9	58775	58083	100530.03	13538.12	23.834	6.154	0.420	0.048	24.255	6.202	56.8	57.0

Chapter 5

Discussion

Firstly, the undamped and damping performance of the LCL-filter with a grid-connected 5kW inverter system was verified using MATLAB. The bode plots are depicted in figure 4.1 and 4.2 respectively. The resonance peak at 1866.4 Hz in Figure 4.1 was able to be damped using the series passive damping method of the resistor, R_d connected in series with the filter capacitance, C_f . However, there's a slight peak that can be permanently suppressed using complex damping methods discussed by [2], but this is sufficient for our design as it does not affect the overall frequency response. The damping resistor was varied by the changing the filter capacitance to make sure that the system has the same resonating point to ensure system stability [10]. The values of the damping resistor decreases from $R_d = 5.66\Omega$ to $R_d = 1.89\Omega$ when the capacitor was increased from $5.01\mu F$ to $15.01\mu F$. This satisfies the inversely proportional relationship of R_d and C_f .

The resonating point shown in Figure 4.1 satisfies the following inequality for a valid LCL-filter design as discussed in the theory in chapter 2: $10f_g < f_{res} < 0.5f_{sw}$. It can be noticed that all the design cases almost the same attenuation of -60dB/decade. Thus,

The developed variable filter design with design cases showing different combinations of the attenuation percentages taken by L_i and L_g was verified using the MATLAB Simulink model to ensure that the THD is within the limits specified by IEEE519. The model controller was simulated at $f_s = 2000kHz$ which is equivalent to $t_s = 50\mu s$. It can be seen from Figure 4.3 to Figure 4.13 that all the design cases have a THD that is less than 5% as required by IEEE519. However as the attenuation percentage of L_i reduces and the attenuation percentage of L_g increases, the THD becomes slightly better from THD = 3.35% to THD = 3.20%. Design case 4 has a better THD of 3.16% and its inverter-side and grid-side currents are depicted in figure 4.6 and 4.7. These figures show

that the inverter-side THD of the grid is higher than in grid-side but still less than 5% THD limit. Even though, the plots shows a poor attenuation of the switching harmonics compared to other reviewed methodologies, the IEEE519 constraints are still met.

The most important results in this project are shown in Table 4.1. The magnetics inductor design tool was used to generate these results because of how tedious the calculations of power losses are. The magnetic material used for design is the High-flux powder core and the wire size AWG = 12 was used for each design case to handle the 10.25A rated current in the inductors. The core volume, copper loss, core loss, total loss and temperature rise are ranked from 1-9 whereby 1 represent the best design case for that property.

In the investigation, it was found that:

1. Design case 7 where L_i and L_g has 20% and 80% attenuation percentages results in minimum core volume ($79437.63mm^3$).
2. Design case 9 where L_i and L_g has 10% and 90% attenuation percentages results in minimum copper ($29.988W$) and core losses ($0.468W$).
3. Design case 7 where L_i and L_g has 20% and 80% attenuation percentages results in minimum total power loss ($27.259W$).
4. Design case 8 where L_i and L_g has 15% and 85% attenuation percentages results in a small temperature rise ($71.4deg.Celsius$).

Chapter 6

Conclusions

The inverter-side and grid-side inductances (L_i and L_g) can be varied by changing the maximum current ripple and the grid-side current attenuation coefficients. This is used to achieve different attenuation percentages for L_i and L_g *i.e* 50% attenuation in each inductor. The design cases were all found to satisfy the IEEE519 requirement ($< 5\%$ in THD). By integrating magnetic properties, the optimum design cases were found that results in low inductor volume, low losses and better temperature rise. Namely these are the design cases 7, 9 and 8 respectively.

Chapter 7

Recommendations

The developed design methodology for the variable filter design need to be tested on a real hardware to verify the performance of each design case.

Use the IEEE numbered reference style for referencing your work as shown in your thesis guidelines. Please remember that the majority of your referenced work should be from journal articles, technical reports and books not online sources such as Wikipedia.

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Appendix A

Additional Files and Schematics

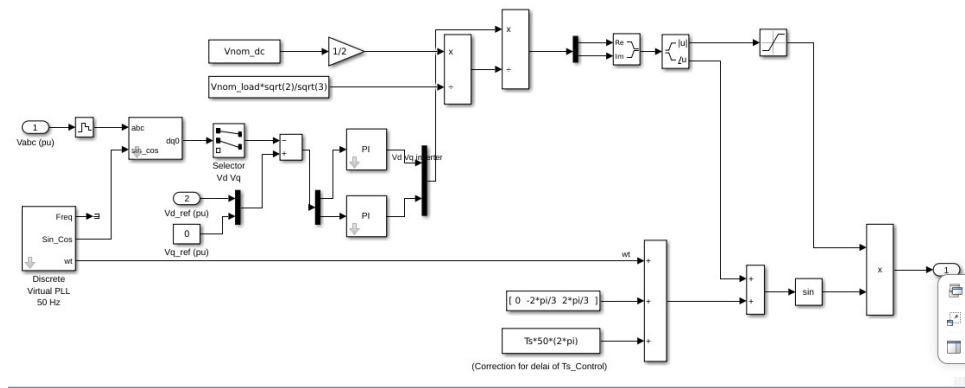


Figure A.1: Control block with the phase-locked loop(PLL) to synchronise the grid voltage

```
% =====Variable LCL Filter Design ===== %
```

```
%%
```

```
clearvars;
```

```
clc;
```

```
%% Given parameters
```

```
s = tf('s');
```

```
Vg = 230;
```

```
U = 230*1.732; % line-to-line RMS voltage
```

```
Fg = 50; % Fundamental grid frequency
```

```
Pr = 5e3; % 5kW Rated active power
```

```
Udc = 480; % DC link voltage
```

```
fsw = 10000; % Switching frequency
```

```
wg = 2*pi*Fg;
```

```

w_sw = 2*pi*fsw;
H = 1/Fg*2; % DC link stored energy constant(s) = 2
cycles
Clink = Pr*H*2/Udc^2; % Link capacitor
%% Derived paramters
Zb = U^2/Pr; % Base Impedance in Ohm
Cb = 1/(wg*Zb); % Base Capacitance in F
%Cb = 100.29e-6;
Ir = (Pr*sqrt(2))/(3*Vg); %Rated current

rip = 0.1596; % Vary the current ripple(rip) allowed to get variable L1
L1 = Udc/(6*fsw*rip*Ir); %inverter-side inductance
ZL = L1 * wg; %inverter-side impedance
Z_perc = ZL/Zb*100;

Ka = 0.0324; %Vary the grid-side current attenuation coefficient Ka to get
variable L2
Cf = 15.01e-6; %Change Cf to get same resonance frequency
Cf1 = Cf;
L2 = (sqrt(1/Ka^2)+1)/(Cf*w_sw^2); % grid-side inductance
r = L2/L1; % Ratio of the grid-side to inverter-side inductance
Xc = 1/(2*pi*50*Cf1);
Xc_b_swit = 1/(2*pi*fsw*Cf1);
Qc = U^2/Xc; % kvar

% Resonance frequency - 10fs<Freq<0.5fsw for unipolar modulation

wres = sqrt((L1+L2)/(L1*L2*Cf1));
fres = wres/(2*pi);
wres = 2*pi*fres;

Rf = 1/(3*wres*Cf); %Damping Resistor

%% Transfer functions

R1 = ZL; %inverter-side parasitic resistance
R2 = (L2*314.159)/5; %grid-side parasitic resistance

% Transfer function of LCL Filter

```

```

Zc = Rf+1/(s*Cf1);
Zl = R1+s*L1;

G_LC = Zc/(Zc+Zl);
f = 1/(2*pi*sqrt(L1*Cf1));

Q = 1/R1*sqrt(L1/Cf1);
damping = 1/(2*Q);
% input Uinv (Inverter) and output at Capacitor (Ucap)

% Current loop LCL with damping resistance
Gi = (s*Cf1*Rf+1)/(s^3*L2*L1*Cf1+s^2*Rf*(L1+L2)*Cf1+s*(L2+L1));
% TF from Uinv to I output (I output is output current in L2 of LCL filter)
% input is Uinv and output is current of L2

%% Current control tuning using PI controller (The current is measured at
(LCL) L2)

C = pidstd(1,1);
C = pidtune(Gi,C,100*2*pi);
Gi_op = C*Gi; % open loop
Gi_fb = feedback(C*Gi,1); % closed loop
% closed loop BW 138 Hz

%% Using sisotool

Gpi = 29.662*((1+0.022*s)/s); % PI tuned using sisotool
Gi_op_sstool = Gpi*Gi;
Gi_fb_sstool = feedback(Gi_op_sstool,1);
% closed loop BW 161 Hz

```

```

%Script for bode plots of the design cases without damping

s = tf('s');

%Design Case 1
l1 = 2.928e-3; %inverter-side inductor
l2 = 2.928e-3; %grid-side inductor

%Design Case 2

```

```

13 = 3.1e-3; %inverter-side inductor
14 = 2.5e-3; %grid-side inductor

%Design Case 3
15 = 3.26e-3; %inverter-side inductor
16 = 2.1e-3; %grid-side inductor

%Design Case 3
17 = 3.452e-3; %inverter-side inductor
18 = 1.8e-3; %grid-side inductor

%Design Case 3
19 = 3.668e-3; %inverter-side inductor
110 = 1.5e-3; %grid-side inductor

%Design Case 3
111 = 3.912e-3; %inverter-side inductor
112 = 1.3e-3; %grid-side inductor

%Design Case 3
113 = 4.191e-3; %inverter-side inductor
114 = 1.0e-3;

%Design Case 3
115 = 4.514e-3; %inverter-side inductor
116 = 0.78e-3; %grid-side inductor

%Design Case 3
117 = 4.89e-3; %inverter-side inductor
118 = 0.54e-3; %grid-side inductor


Cf1 = 5.0145e-6; % Design case 1 capacitor
Cf2 = 5.319e-6; % Design case 2 capacitor
Cf3 = 5.675e-6; % Design case 3 capacitor
Cf4 = 6.13e-6; % Design case 4 capacitor
Cf5 = 6.736e-6; % Design case 5 capacitor
Cf6 = 7.574e-6; % Design case 6 capacitor
Cf7 = 8.834e-6; % Design case 7 capacitor
Cf8 = 10.910e-6; % Design case 8 capacitor
Cf9 = 15.01e-6; % Design case 9 capacitor

```

```

TFA = (1)/((11*Cf1*l2*s^3)+(11+12)*s); % Design case 1 transfer func
TFB = (1)/((13*Cf2*l4*s^3)+(13+14)*s); % Design case 2 transfer func
TFC = (1)/((15*Cf3*l6*s^3)+(15+16)*s); % Design case 3 transfer func
TFD = (1)/((17*Cf4*l8*s^3)+(17+18)*s); % Design case 4 transfer func
TFE = (1)/((19*Cf5*l10*s^3)+(19+110)*s); % Design case 5 transfer func
TFF = (1)/((111*Cf6*l12*s^3)+(111+112)*s); % Design case 6 transfer func
TFG = (1)/((113*Cf7*l14*s^3)+(113+114)*s); % Design case 7 transfer func
TFH = (1)/((115*Cf8*l16*s^3)+(115+116)*s); % Design case 8 transfer func
TFI = (1)/((117*Cf9*l18*s^3)+(117+118)*s); % Design case 9 transfer func

```

hold on

```
% Bode plots for the design cases
```

```

bode(TFA);
bode(TFB);
bode(TFC);
bode(TFD);
bode(TFE);
bode(TFF);
bode(TFG);
bode(TFH);
bode(TFI);

```

hold off

```
%Script for bode plots of the design cases with damping
```

```
s = tf('s');
```

```
%Design Case 1
```

```
l1 = 2.928e-3; %inverter-side inductor
```

```
l2 = 2.928e-3; %grid-side inductor
```

```
%Design Case 2
```

```
l3 = 3.1e-3; %inverter-side inductor
```

```
l4 = 2.5e-3; %grid-side inductor
```

```
%Design Case 3
```

```
l5 = 3.26e-3; %inverter-side inductor
```

```
l6 = 2.1e-3; %grid-side inductor
```

```
%Design Case 3
```

```

17 = 3.452e-3; %inverter-side inductor
18 =1.8e-3; %grid-side inductor

%Design Case 3
19 = 3.668e-3; %inverter-side inductor
110 = 1.5e-3; %grid-side inductor

%Design Case 3
111 = 3.912e-3; %inverter-side inductor
112 = 1.3e-3; %grid-side inductor

%Design Case 3
113 = 4.191e-3; %inverter-side inductor
114 = 1.0e-3;

%Design Case 3
115 = 4.514e-3; %inverter-side inductor
116 = 0.78e-3; %grid-side inductor

%Design Case 3
117 = 4.89e-3; %inverter-side inductor
118 = 0.54e-3; %grid-side inductor


Cf1 = 5.0145e-6; % Design case 1 capacitor
Cf2 = 5.319e-6; % Design case 2 capacitor
Cf3 = 5.675e-6; % Design case 3 capacitor
Cf4 = 6.13e-6; % Design case 4 capacitor
Cf5 = 6.736e-6; % Design case 5 capacitor
Cf6 = 7.574e-6; % Design case 6 capacitor
Cf7 = 8.834e-6; % Design case 7 capacitor
Cf8 = 10.910e-6; % Design case 8 capacitor
Cf9 = 15.01e-6; % Design case 9 capacitor

%Damping Resistors

Rf1 = 5.66; %Case 1
Rf2 = 5.34; %Case 2
Rf3 = 5.00; %Case 3
Rf4 = 4.64; %Case 4
Rf5 = 4.22; %Case 5

```

```

Rf6 = 3.75; %Case 6
Rf7 = 3.22; %Case 7
Rf8 = 2.61; %Case 8
Rf9 = 1.89; %Case 9

TFA = (1)/((l1*Cf1*l2*s^3)+Cf1*(l1+l2)*Rf1*s^2+(l1+l2)*s); % Design case 1
      transfer func
TFB = (1)/((l3*Cf2*l4*s^3)+Cf2*(l3+l4)*Rf2*s^2+(l3+l4)*s); % Design case 2
      transfer func
TFC = (1)/((l5*Cf3*l6*s^3)+Cf3*(l5+l6)*Rf3*s^2+(l5+l6)*s); % Design case 3
      transfer func
TFD = (1)/((l7*Cf4*l8*s^3)+Cf4*(l7+l8)*Rf4*s^2+(l7+l8)*s); % Design case 4
      transfer func
TFE = (1)/((l9*Cf5*l10*s^3)+Cf5*(l9+l10)*Rf5*s^2+(l9+l10)*s); % Design case 5
      transfer func
TFF = (1)/((l11*Cf6*l12*s^3)+Cf6*(l11+l12)*Rf6*s^2+(l11+l12)*s); % Design
      case 6 transfer func
TFG = (1)/((l13*Cf7*l14*s^3)+Cf7*(l13+l14)*Rf7*s^2+(l13+l14)*s); % Design
      case 7 transfer func
TFH = (1)/((l15*Cf8*l16*s^3)+Cf8*(l15+l16)*Rf8*s^2+(l15+l16)*s); % Design
      case 8 transfer func
TFI = (1)/((l17*Cf9*l18*s^3)+Cf9*(l17+l18)*Rf9*s^2+(l17+l18)*s); % Design
      case 9 transfer func
hold on

% Bode plots for the design cases
bode(TFA);
bode(TFB);
bode(TFC);
bode(TFD);
bode(TFE);
bode(TFF);
bode(TFG);
bode(TFH);
bode(TFI);
hold off

```
